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Policy, regulation, development and future of biodiesel industry in Brazil

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ABSTRACT

This article provides a detailed analysis of the Brazilian biodiesel market and policies over the past 16 years. The biodiesel market faced distinct periods, from the initial non-mandatory 2% to the actual 13% mixture in diesel.

The legal and regulatory basis of the Biodiesel Program, as well as the tax benefits to poorer regions and family farmers, was also explained. The biodiesel industry increased from almost no biodiesel production in 2005 to more than 4 billion liters at the end of 2017.

Soybean oil is the major raw material in Brazilian biodiesel (around 90%), although fats are also important. Other materials, such as sunflower oil, peanut oil, castor oil, palm oil (mainly in North and Northeast regions) and used oil (especially in the Southeast region) showed increased participation.

Biodiesel content could reach 15% in 2023. That increase can lead to an expansion in biodiesel demand to 9 billion liters in 2024, which will need soy processing improvement, in order not to take place at the expense of an increase in deforestation and smallholder agriculture. Some recent events, such as the reduction of oil prices and the pandemic of Covid-19 can diminish the rate of that expansion.

1. Introduction

Despite that the possibility to use vegetable oils in combustion engines (Shay, 1993), is known since 1900, the low cost and the large availability of fossil fuels remained the major energy source, responsible for one-third of the world primary energy consumption in 2016 (BP Statistical Review of World Energy, 2018). Due to periods when fossil fuel prices increase, such as the petroleum international crises in the 1970s and 1990s, the search for other energy sources increased. Zanin et al. (2000) studied the bioethanol program, with sugarcane as the main raw material, while Parente (2003) investigated the biodiesel industry, its raw materials, industrial processes as well as economic, social and environmental aspects. Detailed review on transesterification of vegetable oil with a focus on the process (Schuchardt et al., 1998) and on the raw materials (Hill, 2000). Later, the possibility of substitution of oil by renewable materials in the petrochemical industry was studied (Schuchardt et al., 2001).

In Brazil, similar behavior can be observed, with the creation of the successful PROALCOOL Program (with a focus on ethanol) in the 1970s and the Brazilian Production and Use of Biodiesel Program (PNPB) by the 1990s.

Biodiesel is a renewable fuel, described as a mixture of monoalkyl esters, especially methyl esters, produced via the transesterification process, which is the reaction of vegetable oil (or animal fat) with an alcohol (methanol or ethanol) in the presence of a catalyst (usually alkali, due to better results), producing biodiesel and glycerol. Due to the increase of glycerol production, studies on its use in renewable energy generation through processes such as fermentation, digestion, gasification, pyrolysis, liquefaction, combustion, and steam reforming (He et al., 2017), as well as the manufacture of chemical products (Monteiro et al., 2018). Biodiesel exhibits lower viscosity and oxygen content when compared to vegetable oil. Since methanol is less expensive than ethanol and other superior alcohols, it is largely used. Biodiesel is also biodegradable and non-toxic (Demirbas, 2011), sulfur-free (prevents acid rain), decreases soot emission (reducing respiratory diseases and health expenses), is almost neutral about CO_2 emission (Parente, 2003) and acts as engine lubricant (increasing engine lifetime).

Although PNPB aims at oilseed diversification, Alves et al. (2017) showed that the used mechanisms had little impact on biodiesel feedstock diversification, with the predominance of biodiesel production from soybean oil.

Some literature was produced in the first years of the PNPB Program. Pousa et al. studied the history and policy of biodiesel (Pousa et al., 2007), while Sorda et al. (2010) studied both biodiesel and bioethanol, comparing Brazilian data to other countries and, Zonin et al. focus on a case study (Zonin et al., 2014), as well as some deepening in issues such as social (Oliveira et al., 2019), food security (Finco and Doppler, 2010), environmental sustainability (Garcez and Vianna, 2009), land use and

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its expansion (Castanheira et al., 2014), agricultural impacts on palm culture (Cesar and Batalha, 2013) and castor oil (Conejero et al., 2017) and alternative biomass (Bergmann et al., 2013). A recent article deals with the evolution and environmental impacts of the biodiesel program in Brazil until 2015 (Oliveira and Coelho, 2017).

The objective of this paper is to provide an overview of the Brazilian biodiesel market and policies through the past 16 years from the regulator point of view. The biodiesel market faced distinct periods, from the non-mandatory one to 8% on diesel fuel as well as the present 13%. This article also presents the key laws and regulations, as well as some of the incentive schemes used and operational rules. We also identify the feedstock employed and discuss the impacts of the PNPB program on diesel imports. The article also proposes a strategy for the implementation of the biodiesel industry, to help other countries not to face some wrong decisions made by Brazil.

2. National program of production and use of biodiesel (PNPB)

Brazilian Production and Use of Biodiesel Program (PNPB), published in July 2003 (Brazil, 2003), launched the goals which made possible the insertion of small, medium and large companies from different industrial segments, such as fertilizers, cosmetics, alcohol, among others, in the biodiesel production activity. That makes this industry extremely heterogeneous, with different operational performances and different behavior regarding occupational safety and health as well as environmental protection.

At that time, a Presidential Decree created the Interministerial Working Group on Biodiesel (GTIB - Grupo de Trabalho Interministerial: Biodiesel), to study and point the viability of using biodiesel as an alternative source of energy (Brazil, 2003). The preference for vegetable oil indicates the inclination to improve agriculture in Brazil, mainly in poorer states. The use of residual oils (of vegetable or animal origin) was not mentioned on GTIB, although they are a profitable alternative. Nelson and Schrock study was focused on beef tallow (Nelson and Schrock, 2006) while Tsai et al. used waste edible oil as the biodiesel raw material (Tsai et al., 2007).

The report of GTIB indicated the viability to introduce biodiesel into the countries energy matrix and launch the basis to Brazilian policy for biodiesel (Rodrigues, 2003) with emphasis on the possibility of reducing emissions on diesel engines.

During 2004, the Interministerial Executive Commission (under Civil House) and the Biodiesel Managing Group (coordinated by the Ministry of Mines and Energy), was in charge to study and implement a biodiesel regulation basis. That regulation includes biodiesel percent, tributes structure, quality standards, and economics of the new market.

Yet in 2004, the Ministry of Agrarian Development (MDA) started the elaboration of the basis of the Social Fuel Certification, the Federal Revenue Service defined the incidence of tax incentives and National Development Bank (BNDES) organized a support program for biodiesel producers.

The regulatory basis was given by some Decrees, introducing the biodiesel on the energy matrix. The legal basis on biodiesel production and use came through Law 11097, from January 13rd, 2005 (Brazil, 2005), which establishes a minimum percentage of biodiesel in the blend with diesel fuel, besides the follow-up of the new fuel's insertion in the market and defines ANP (National Petroleum, Natural Gas and Biofuels Agency) as the institution in charge of regulation of biodiesel. At that moment, biodiesel was officially introduced in the Brazilian energy matrix. The legal and regulatory basis take into consideration the use of all types of available triglycerides, the concern with the guarantee of the fuel supply respecting our standards, its competitiveness in comparison to other fuels and our social inclusion policy. That last is one of the PNPB goals, which is to induce familiar grain production, motivating the rural population to stay in rural areas, instead of migrating to urban ones.

That Law fixed in 2% v/v the minimum biodiesel on diesel in January 2008, increasing to 5% until January 2013, which was later modified by National Energy Policy Council (CNPE).

2.1. Social fuel certification and special taxation

The Social Fuel Certification was created by Decree No. 5297, dated December 6, 2004 (Brazil, 2004). This certification was created to promote social inclusion of family farmers taking part in the National Program to strengthen Family Farmers (Programa Nacional de Fortalecimento da Agricultura Familiar – PRONAF), by purchasing a minimum amount of raw materials, within transactions that guarantee incomes and conditions adequate to the activities and to assure technical assistance and capacitation to family farmers. From 2004 to nowadays, some reviews were made due to reorganize the consumption of raw materials and production.

The MDA Normative Instruction No 2, dated September 30, 2005 (MDA, 2005), first determined the minimum percent of raw materials acquisition from family farmers as:

I - 50% (fifteen percent) for acquisitions from North and Semiarid regions;

II - 30% (thirty percent) for acquisitions from Southeast and South regions;

III - 10% (forty percent) for acquisitions from region Northeast and Center-West.

Those percent evolved due to the market improvement, the greater regional demands and agricultural factors, being the last published in 2017.

The MDA Ministerial Order No 512, dated September 5, 2017 (MDA, 2017), brings the actual proceedings and criteria to grant the Social Fuel Certification, which includes a minimum percent of raw materials acquisition from family farmers of:

I - 15% (fifteen percent) for acquisitions from North and Center-West;

II - 30% (thirty percent) for acquisitions from Southeast, Northeast, and Semiarid;

III - 40% (forty percent) for acquisitions from the South region.

That minimum percent is calculated based on the annual cost of raw material from family farmers against the annual cost of raw material used in the biodiesel plant.

The biodiesel producer granted the Social Fuel Certification, received access to significantly lower PIS/Pasep (Social Integration Program and Public Servant Estate Fortification Program) and COFINS (Social Security Financing Contribution) values.

That tax arrangement considers not only that biodiesel is a renewable fuel, so it must have low federal taxes, but also that some regions or raw materials must have some improved support. Decree No. 5297, dated December 6, 2004 (Brazil, 2004) created four different categories: i. palm oil and castor oil from the north, northeast, and semiarid regions, from family farmers; ii. Other raw materials, from family farmers; iii. palm oil and castor oil from the north, northeast, and semiarid regions; iv. Others.

The program has the intent to develop arrangements with family farmers, preferentially in the poorest regions of the country (north, northeast, and semiarid regions – most of the northeast region and the north of Minas Gerais state) and with incentives to castor oil and palm oil. The applied values are summarized in Table 1.

In 2008, Decrees 6458 and 6606 (Brazil, 2008a, 2008b) altered the category "palm oil and castor oil from the north, northeast and semiarid regions, from family farmers", removing the limitation on palm oil and castor oil. At that time, the poorer regions still need support to develop that economic activity, but castor oil showed to be a not so good raw material (due to high price and limitations on biodiesel quality parameters) and palm oil did not increase its production as expected (due mainly to its wide production cycle). Some reduction in the tax value on general biodiesel showed an improvement in the support for the Program.

Table 1
Tax values on biodiesel (Brazil, 2004) (R$ per cubic meter).

Tax values	PIS[a]	COFINS[a]
All other biodiesel (general biodiesel)	39.65 (14.59)	182.55 (67,18)
Palm oil and castor oil from the north, northeast, and semiarid regions	27.03 (9.94)	124.47 (45.81)
Other raw materials (besides castor and palm oil), from family farmers	12.49 (4.59)	57.53 (21.17)
Palm oil and castor oil from the north, northeast, and semiarid regions, from family farmers	0	0

[a] Values between parenthesis are in US$ per cubic meter, calculated from currency exchange between Real and US dollar at the time of the publication of the Decree.

That change in Social Fuel Certification was better discussed by Rathmann et al. (2012) and Oliveira et al. (2019). The former showed an early view, while the last brought some problems during its implementation. As discussed, the production of castor and palm oil was expected to enhance small farmers from the North and Northeast economy. They were sent to other more profitable markets, like cosmetics, pharmaceutical, and culinary markets. They also have not reached a large scale to meet market demand. Other difficulties of these oils are related to meeting the quality requirements (castor oil), its micro pulverized production (increasing production and transport costs), low utilization of agricultural machinery and dependence on artificial irrigation.

The support to north, northeast and semiarid regions can be understood by the observation of data in Table 2. Data from the year 2000 shows that north and northeast regions have the lower Municipal Human Development Index (MHDI). It is worth mentioning that values from 0,500 to 0,599 are considered low, as well as from 0,600 to 0,699 are medium. In 2000, only Distrito Federal and São Paulo showed high development index (from 0,700 to 0,799).

It is impossible to measure what was the contribution of PNPB to the development of any region or state, but it is evident that development policies to the mentioned regions were successful. MDHI from north and northeast regions goes from 0,541 to 0,512 (data from 2000) to 0,684 and 0,660 (data from 2000), indicating an increase of more than 25%, while other regions faced an increase of around 15%. It is worth mentioning that in 2010, a greater number of states reached a high development index.

Table 2
Municipal Human Development Index for years 1991, 2000 and 2010 (UNDP et al., 2013).

Regions and states	MHDI 1991	MHDI 2000	MHDI 2010
Midwest	0.510	0.639	0.753
Distrito Federal	0.616	0.725	0.824
Goiás	0.487	0.615	0.735
Mato Grosso	0.449	0.601	0.725
Mato Grosso do Sul	0.488	0.613	0.729
North	0.422	0.541	0.684
Acre	0.402	0,517	0.663
Amapá	0.472	0.577	0.708
Amazonas	0.430	0.515	0.674
Pará	0.413	0.518	0.646
Rondônia	0.407	0.537	0.690
Roraima	0.459	0.598	0.707
Tocantins	0.369	0.525	0.699
Northeast	0.393	0.512	0.660
Alagoas	0.370	0.471	0.631
Bahia	0.386	0.512	0.660
Ceará	0.405	0.541	0.682
Maranhão	0.357	0.476	0.639
Paraíba	0.382	0.506	0.658
Pernambuco	0.440	0.544	0.673
Piauí	0.362	0.484	0.646
Rio Grande do Norte	0.428	0.552	0.684
Sergipe	0.408	0.518	0.665
South	0.531	0.663	0.756
Paraná	0.507	0.650	0.749
Rio Grande do Sul	0.542	0.664	0.746
Santa Catarina	0.543	0.674	0.774
Southeast	0.534	0.658	0.754
Espírito Santo	0.505	0.640	0.740
Minas Gerais	0.478	0.624	0.731
Rio de Janeiro	0.573	0.664	0.761
São Paulo	0.578	0.702	0.783

In 2012, Decree 7768, dated June 27, 2012 (MME, 2007) reduced all tax values. These are the current ones (Table 3).

Besides those benefits, biodiesel producers having Social Fuel Certification can also access special financing products at National Social and Economic Development Bank (BNDES) and its financial accredited institutions, among others.

2.2. Biodiesel auction

To stimulate the biodiesel market before the beginning of the mandatory period (during the facultative period), biodiesel auctions were developed, under the responsibility of ANP, where the fossil diesel producer and importer (mainly PETROBRAS) ensures to purchase of produced biodiesel. The MME (Ministry of Mines and Energy) Ministerial Order No 284, dated October 4, 2007 (MME, 2007) and Resolution CNPE No 5, dated October 5, 2007 (CNPE, 2007), defined biodiesel auction guidelines, such as i. 80% of biodiesel to be marketed should be from biodiesel producers with social fuel Certification; ii. All biodiesel must fulfill ANP quality requirements; iii. All biodiesel must be produced by the seller and iv. ANP must determine the penalties to whom do not produce and deliver biodiesel according to the rules of the auction.

It is worth knowing that before the 2% mandatory period, five auctions were done, to buy 885 million liters of biodiesel, which was used from January 2006 to December 2007. That biodiesel was commercialized at prices from R$ 1750.00 to R$ 1900.00, per cubic meter.

The auction to comply with the mandatory 2%, from January 2008 to June 2008 was done in November 2007. At that auction, around 30 biodiesel producers competed to sell 380.000 m^3. The cubic meter was sold for around R$ 1865.00. At that time, 80% of the volume came from producers with social fuel Certification.

The World Crisis in 2008 had a strong influence on the prices of biodiesel this year. The exchange rate between currencies (real against the dollar) changed significantly, which led the biodiesel to prices 40% higher than the last ones, reaching more than R$ 2600.00, per cubic meter Fig. 1 shows the average prices (in R$/$m^3$) (ANP, 2018a). At that time, biodiesel content also became to rise (Fig. 2), with the increase in biodiesel demand.

Besides that, the fast increase in biodiesel content on fossil diesel, from 2 to 5% was another reason for the high values negotiated for biodiesel, from R$ 2150.00 to R$ 2600.00, per cubic meter.

In the beginning, the biodiesel auctions took place at different periods, since it was a new experience. From August 2008 on, the auctions were performed for 3 months period, until 2012, when the period was

Table 3
Tax values on biodiesel (Brazil, 2012) (R$ per cubic meter).

Tax values	PIS[a]	COFINS[a]
All other biodiesel (general biodiesel)	26.41 (12.03)	121.59 (55.39)
Palm oil and castor oil from the north, northeast, and semiarid regions	22.48 (10.24)	103.51 (47.15)
All raw materials (besides north, northeast, and semiarid regions), from family farmers	10.39 (4.73)	47.85 (21.79)
All raw materials from the north, northeast, and semiarid regions, from family farmers	0	0

[a] Values between parenthesis are in US$ per cubic meter, calculated from currency exchange between Real and US dollar at the time of the publication of the Decree.

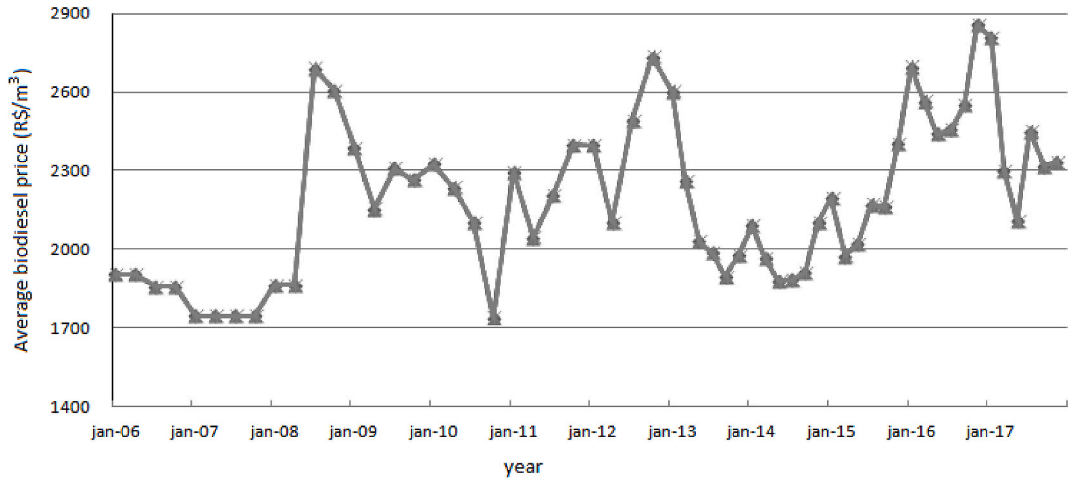

Fig. 1. Average biodiesel price on auctions (ANP, 2018a).

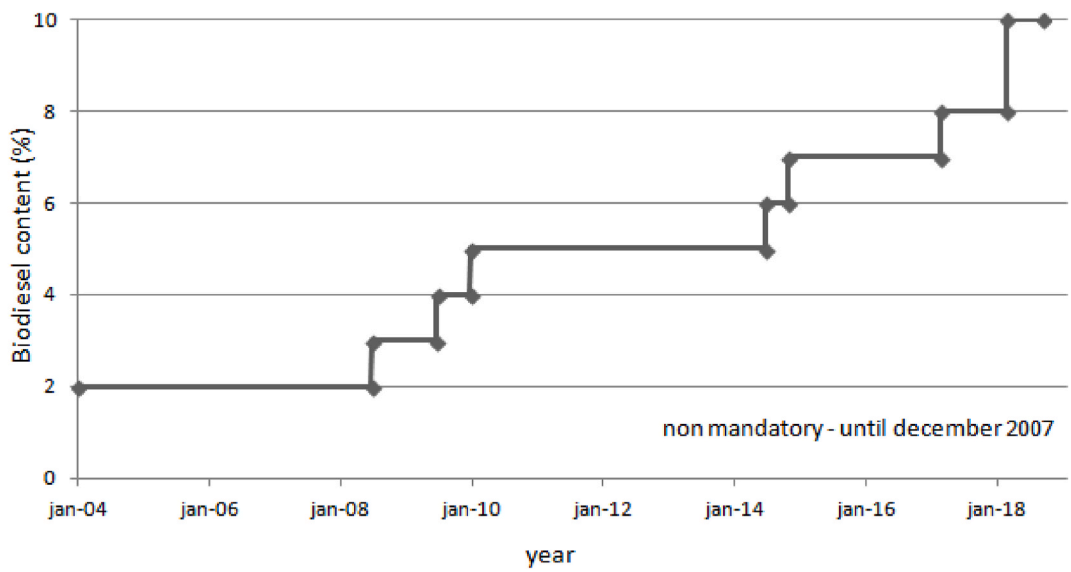

Fig. 2. Biodiesel content on fossil fuel (Brazil, 2005, 2014; CNPE, 2008, 2009a, 2009b, 2016, 2017).

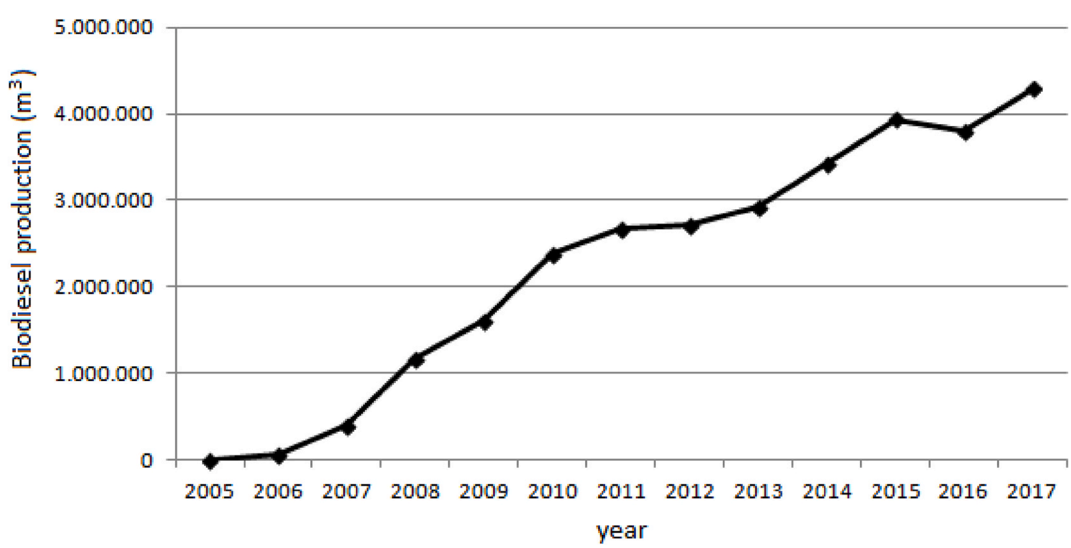

Fig. 3. Biodiesel production (ANP, 2018b)

reduced to 2 months. Since Brazil does not have significant biodiesel exports, biodiesel production is subordinate to sales on auctions. Since the beginning, biodiesel companies complain that a defeat on auctions (available biodiesel capacity is higher than demand) is equivalent to a period without production. So, the shorter the period from one auction to another, the better conditions for the market.

Comparing biodiesel prices against biodiesel content on fossil diesel, it is easy to verify that the price does vary mostly because of dollar and raw materials (mainly soy oil) fluctuations than because of the increase of the biodiesel market. That is also expected, since biodiesel production uses only 40–65% of industrial installed capacity, from 2008 to the actual moment.

3. Production and consumption evolution (2004–2017)

Figs. 3–6 exhibit the biodiesel production, as well as the biodiesel average capacity, the number of biodiesel units and daily authorized capacity, with the percent of the use of it, from 2005 to 2017.

The comparison between all these data will lead to a better understanding of the dynamics of the different periods of the PNPB (2004–2007, 2008–2010, 2010–2014, 2014–2017).

Average authorized capacity (Fig. 4), is an important parameter to understand the development of the biodiesel industry. It is obtained by the ratio of authorized capacity and the number of biodiesel units. So, it indicates the rate of increase and modification of the units, to higher processing capacity, and so, to higher risks.

The biodiesel consumption of more than 4 billion liters at the end of 2017, putting Brazil as the second world biodiesel producer (REN21, 2018), can let us consider PNPB as an achievement of this energy program.

3.1. Between 2004 and 2007 (2% mixture, non-mandatory)

Although biodiesel was stated in the Law as a "biofuel derived from renewable biomass for use in engines of internal combustion with ignition by compression or generation of another type of energy, that can partially or completely substitute fossil fuels'', quality parameters defined by ANP limited the use of biodiesel obtained through trans-esterification or esterification (methyl or ethyl fatty acids' esters), mainly methyl fatty acids' esters.

Regulation related to the production of biodiesel during that period (ANP Resolution 41/2004 from November 24, 2004) was very simple, with no entry barriers. This was adequate to the main goal, which was to induce the construction of biodiesel plants and promote the materialization of a biodiesel industry (ANP, 2004).

Figs. 3–6 evidenced that the objective of building a biodiesel industry was reached, since at this non-mandatory period, showed the most prominent increase of biodiesel plants can be observed, reaching 45 biodiesel units at the end of 2007. In that period, production was also small, which led to the maintenance of high idle capacity, around 83%.

At this time, as already mentioned, the use of biodiesel was not mandatory. Its use, mixed with diesel fuel was carried by a small amount of fuel distribution companies, with the leadership of the Petrobras Distributor. Due to its improvement in the environment, at that time, biodiesel was used even for propaganda among the consumers.

At the end of 2007, biodiesel authorized capacity was 6 times biodiesel consumption. But at that time, the average authorized capacity increased from 30 m^3/day (in 2005) to 153 m^3/day (in 2007). In other words, the new biodiesel units had an average capacity higher than five times the first authorized units. That overcapacity was due to the next expected steps of the program, with a target to reach 5%, in a few years. That indicates that the market was on the way to comply with the demand.

Due to the increase in the biodiesel industry, it became clear that the first ANP regulation on biodiesel plants needs improvement. The more complex the biodiesel units were becoming, the higher the risk of an industrial accident.

3.2. Between 2008 and 2010 (2–5% mixture)

The knowledge acquired during almost 4 years let the country build up a more restrained regulation (ANP Resolution 25/2008 from September 2nd, 2008) (ANP, 2008). That new regulation was based on three steps: To build a biodiesel unit, it was necessary to be granted an Authorization to Construct. That step allows the country to see if the industry's response to the demand for the program will be fulfilled. After the construction, of the industrial facility, the company must have the Authorization to Operate the facility, letting the company start the operation, to produce biodiesel. That Authorization is granted after ANP inspection when it can be verified the use of good engineering practices and the accordance with National regulation. And, finally, the Authorization to Sale, which indicates that biodiesel produced is in harmony with fuel quality standards. Some other improvements on regulation let us consider that this regulation was much more adequate for this new industry than the older one.

At that time, Brazil had 68 (sixty-eight) authorized biodiesel facilities and 19 (nineteen) biodiesel facilities demanding authorization (enlargement, mainly). The biodiesel production and the number of

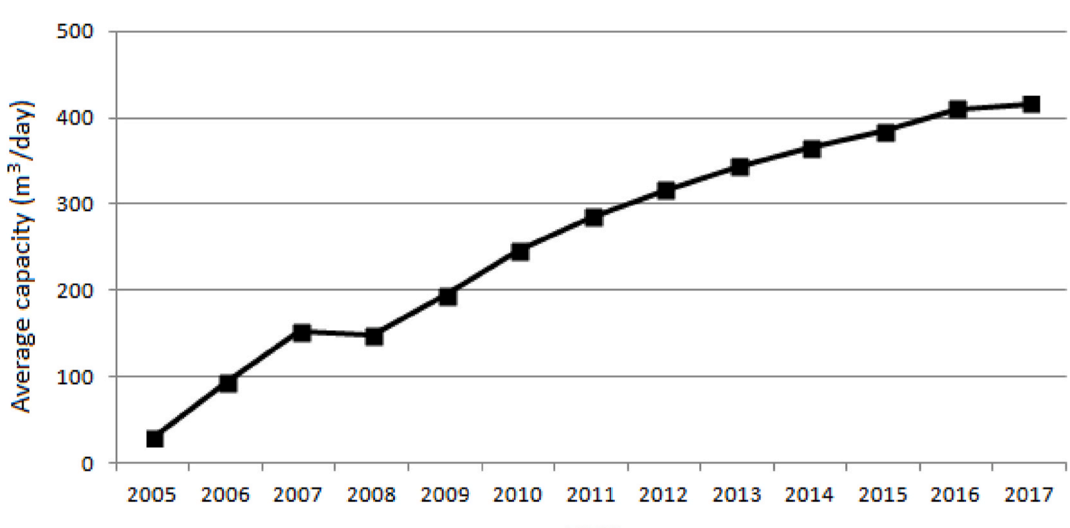

Fig. 4. Biodiesel average capacity (ANP, 2018b).

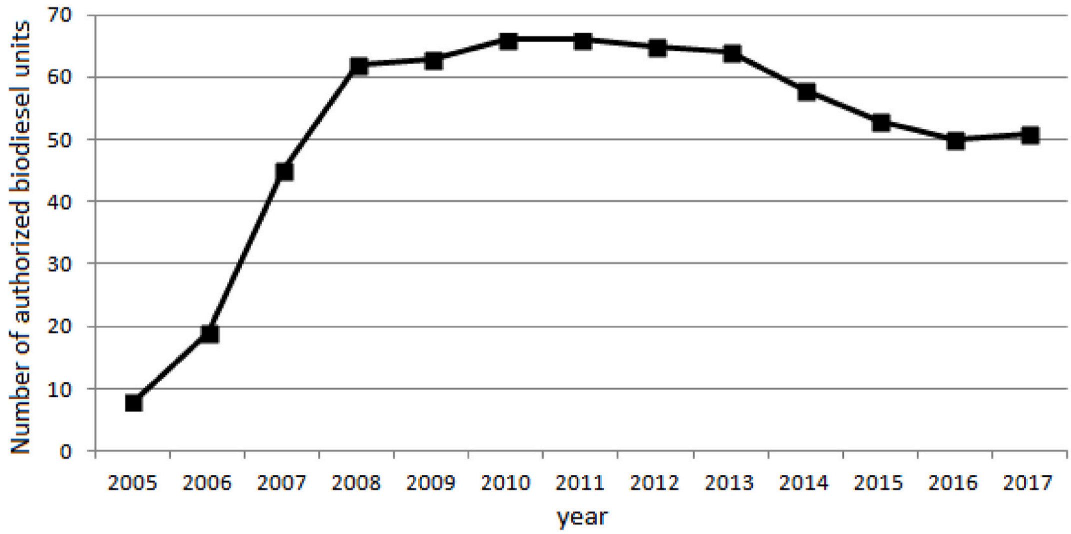

Fig. 5. Number of authorized biodiesel units (ANP, 2018b).

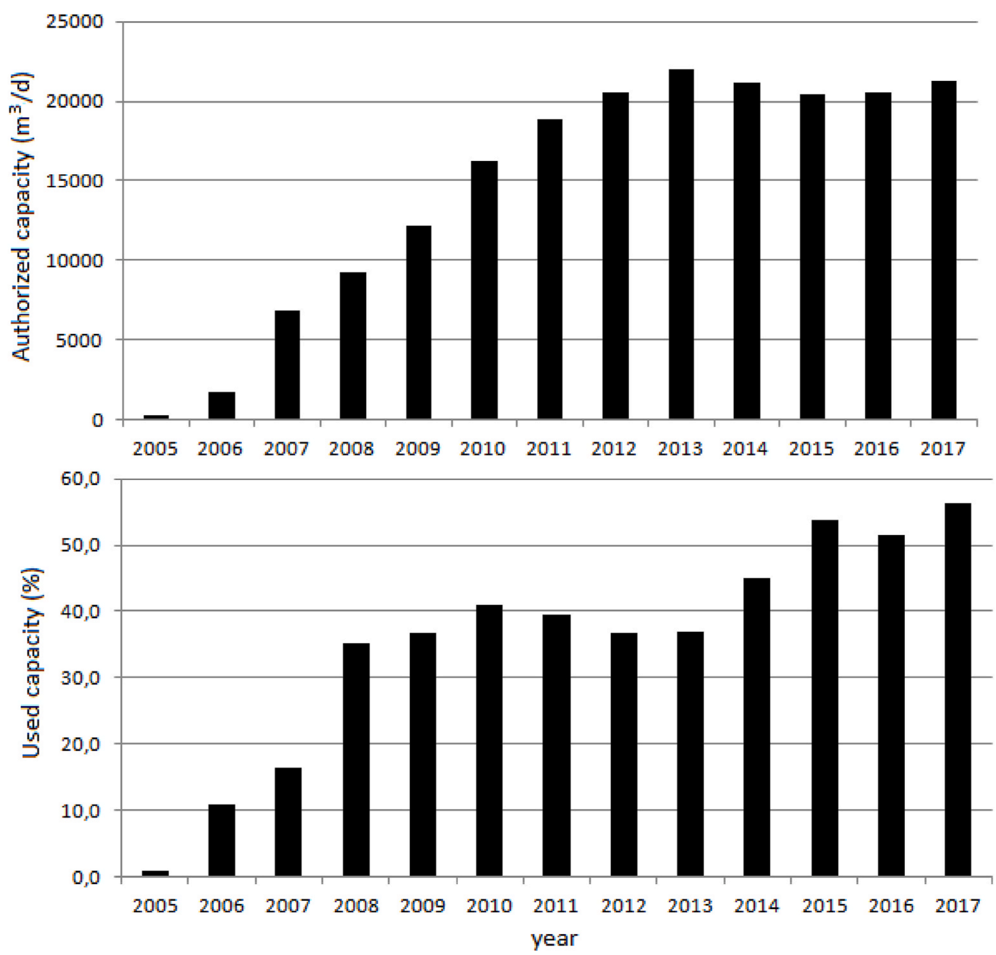

Fig. 6. Daily authorized capacity and percent of the use of industry (ANP, 2018b).

biodiesel facilities rose significantly during the last few years. That remarkable increase was motivated by the increase of the mandatory amount of biodiesel in diesel fuel sold in Brazil.

Biodiesel demand rose over 2 million cubic meters (in 2010) and the used capacity reach over 40%.

Due to the location of most of the plants around vegetable oil production plants, far from its destination, it was also necessary to establish rules for the storage of biodiesel, to organize logistics of production, sale, transport, and use.

That new regulation was well discussed with all players and society and was the result of market perceptions.

ANP regulation also has a different proceeding to the construction of small biodiesel plants (production of 30,000 L a month), when all production is intended to be used for research or self-consumption. Due to ANP regulation, those non-commercial units were released to get ANP authorization, leaving only the commercial ones to fulfill ANP regulation.

Another important event in that period was that some Brazilian cities, mainly in the South region, during the winter, faced filter obstruction due to the use of large amounts of fats on the raw material. That was solved with the limitation of the use of fats on biodiesel produced for cities where the temperature goes to low ones, especially in the mountains.

Since the biodiesel content on diesel fuel increased during that period, production was also developed. Due to that, the number of biodiesel plants increased and reached 66 units at the end of 2010. The average capacity reaches 246 m^3/day, which indicates that bigger plants were operating in Brazil.

3.3. 2010 to 2014 (5% mixture)

The early years of ANP Resolution 25/2008 let us see that some points were not as clear as expected and the biodiesel industry questioned ANP to enlighten some points. In mid-2011 ANP started the discussion on the publication of a new Act to replace the current one implementing actions to improve the regulation of the productive segment of biodiesel. That improvement comes from the experience gained during the time called the “learning curve".

One of the improvements was the introduction of a prescriptive regulation on construction and operation requirements.

Before that, some requirements were also requested during authorization inspection. But although expected by the market, the requirements were not written. Putting it clear on regulation, after the discussion was a request of both the biodiesel industry as from society (transparent requirements, letting everyone knows in advance what is expected from the industrial site). To the full compliance, it was given 18 months after approval and publication (ANP Resolution 30/2013 from September 2nd, 2013) (ANP, 2013).

That period of fixed biodiesel percent, from January 2010 to July 2014, was less dynamic than the other ones. The number of biodiesel units changed from 63 to 58.

Although the number of units showed a small decrease, the average capacity increased from 194 to 356 m^3/day, which indicates that the biodiesel units were improving continuously their capacities.

That period, where the biodiesel demand increased, mainly to diesel increase in demand, since biodiesel content was fixed at 5%. That fact was very important to the behavior in the next period, when a significant decrease of the biodiesel plants, reaching its minimum of 50 units at the end of 2016, as will be discussed on the next issue.

3.4. 2014 to 2017 (increase from 5% to 10%)

As already mentioned, this period was marked by a significant decrease in the number of biodiesel plants, which started slowly in 2011. Although the number of biodiesel plants decreased, the average capacity showed a small increase, which led us to conclude that small plants did not stay viable during the last period, due to a small increase in demand (Fig. 4).

Although the new ANP regulation gave 18 months to the adaptation of the existing plants, some biodiesel units were already out of operation. These units, due to economic issues, usually decided not to adapt their plants and get out of the market. These new regulation requirements allied to the fact that more than 50% of the units were inspected by ANP experts, results in a more efficient and safer biodiesel industry.

Due to the reduction of the number of biodiesel plants and an increase in consumption, the biodiesel industry is almost 55% of its installed capacity.

It is expected an even higher development of that industry during the next years, since it was expected the gradual increase from 5% to 10% of biodiesel in fossil diesel, by 2015, that action was expected to stimulate the construction of new biodiesel facilities, the enlargement of the existing ones, as well as the growth of biodiesel infrastructure. Strong management of the supply chain becomes necessary from the blends of biodiesel, such as B10 or B20.

Besides what was explained, here we have some other challenges, such as the need to observe and understand new biodiesel production technologies (different from transesterification), since they can become important to the fuel market through the next years. Besides the attribution to regulate fuel market, ANP has also the mission to encourage and finance research, to improve biodiesel production, to encourage the study and the use of non-usual raw materials to biodiesel production and study and encourage the prospecting of special industrial uses, such as drilling fluid on the oil industry.

Fig. 7 shows the geographic distribution of biodiesel facilities. It can be seen that they are predominantly located in vegetable oil or animal fat production areas, such as the states of Mato Grosso (MT), Goias (GO), Mato Grosso do Sul (MS) and Rio Grande do Sul (RS). A smaller fraction, but also important, is located close to fuel consumption, such as São Paulo (SP). That is a characteristic of Brazilian biodiesel industry development, where a good number of raw producers became biodiesel producers.

Moreno-Pérez et al. (2017) recently visited some works on the de-concentration of the biodiesel market over the years. Data on auctions show the decreasing of the Herfindahl-Hirschman Index (HHI), going to values close to 0.05 on the last auction of 2017. That indicates that the Brazilian biodiesel industry reached some maturity, with a highly competitive market.

The geographical distribution of biodiesel production units also presents an additional concern for the fuel sector. Since most of the units are in the interior of the country, near the production of oil and fats and the methanol used in the manufacture of biodiesel is largely imported, this methanol travels throughout much of the national territory.

As the price of methanol is on average lower than that of ethanol (Brazil is a major producer), there is a permanent need to oversee this product to prevent fraudsters from mixing methanol in ethanol, as the former has high toxicity.

After biodiesel production, methanol continues to concern the fuel and biofuels sector, since part of the biodiesel production units does not have the equipment to recover the methanol used, but it is not incorporated into the biodiesel. In general, the methanol-containing stream is routed to chemical industries hundreds of miles away, with the risk of fraudsters using it.

It is worth mentioning that at the end of 2017, a new program, named Biofuels National Policy - Renovabio (Brazil, 2017) was launched. Renovabio aims to improve the production and use of biofuels, as well as improve discussion on Climate Change and sustainability. The regulation concerning that policy is in construction.

4. Used raw materials

The diversification of biodiesel feedstock was the main objective of PNPB, but its achievement has remained deceptive.

Alves et al. (2017) indicated that tax reductions and other benefits of PNPB have a positive impact on the profitability of biodiesel production chains of soybean, rapeseed, and sunflower, with the superiority of the soybean industry, due to its efficiency and competitiveness.

The main raw material for the transesterification reaction can be vegetable oil or animal fat. But, due to the biodiesel demand, which goes from almost nothing in 2003 to 4.3 million cubic meters in 2017 and to what was pointed by Alves et al. (2017), raw materials for biodiesel production became very dependent on soybean oil and animal fats.

Although either methanol or ethanol could be used, methanol was the most alcohol used. Exception for one or two biodiesel units, in the

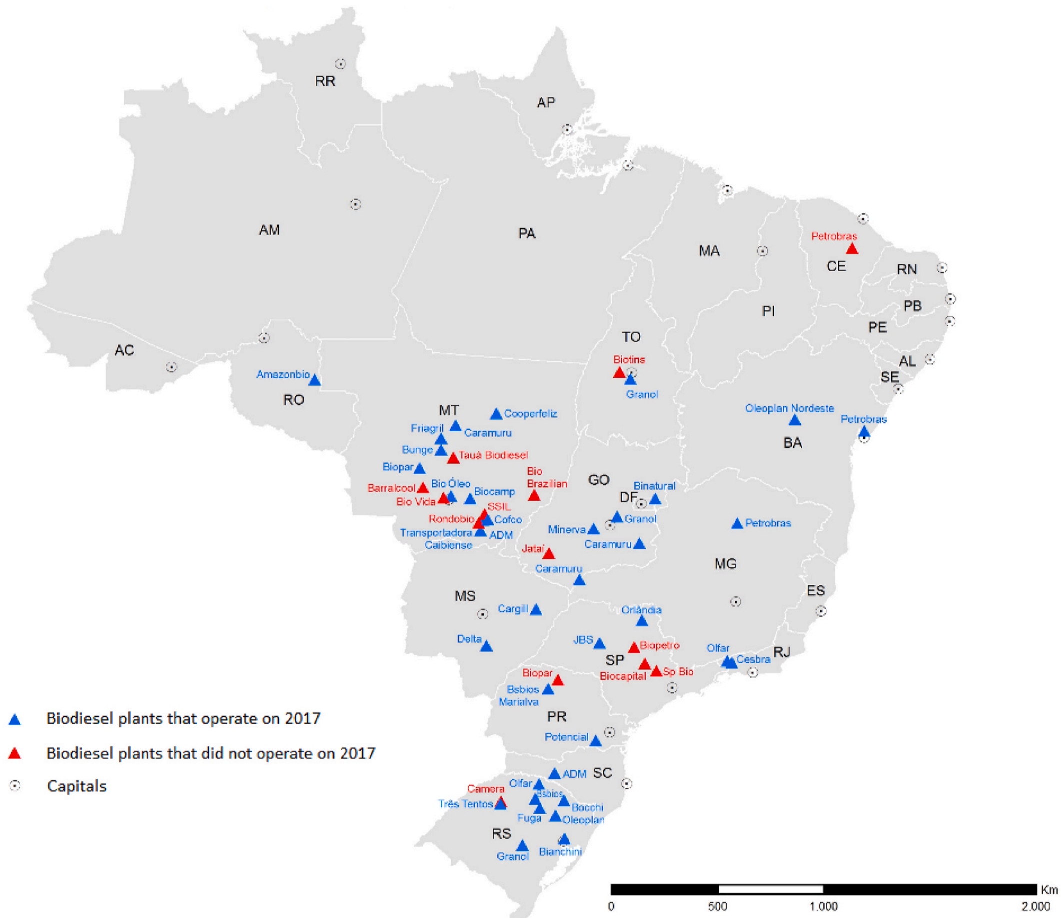

Fig. 7. Geographic distribution of biodiesel facilities (ANP, 2019)

early years of the PNPB, that used ethanol on transesterification reaction. During the last years, methanol had lower prices than ethanol and its conversion to biodiesel is better improved with methanol, Methanol recovery is also less expensive than ethanol one since ethanol shows the formation of a binary azeotrope.

Other used vegetable oils are palm, rapeseed, cotton, and sunflower. In the last few years, some waste oil from cooking, started to be used in the production of biodiesel. Although they can show regional relevance, in a national scenario, they do not. Indeed, soybean oil and animal fats are responsible for more than 90% of the raw used for biodiesel production.

Conejero et al. (2017) analyzed the competitiveness of the castor bean organizational arrangement. They believed that castor beans cannot become competitive to beat others, which they ascribed to weak

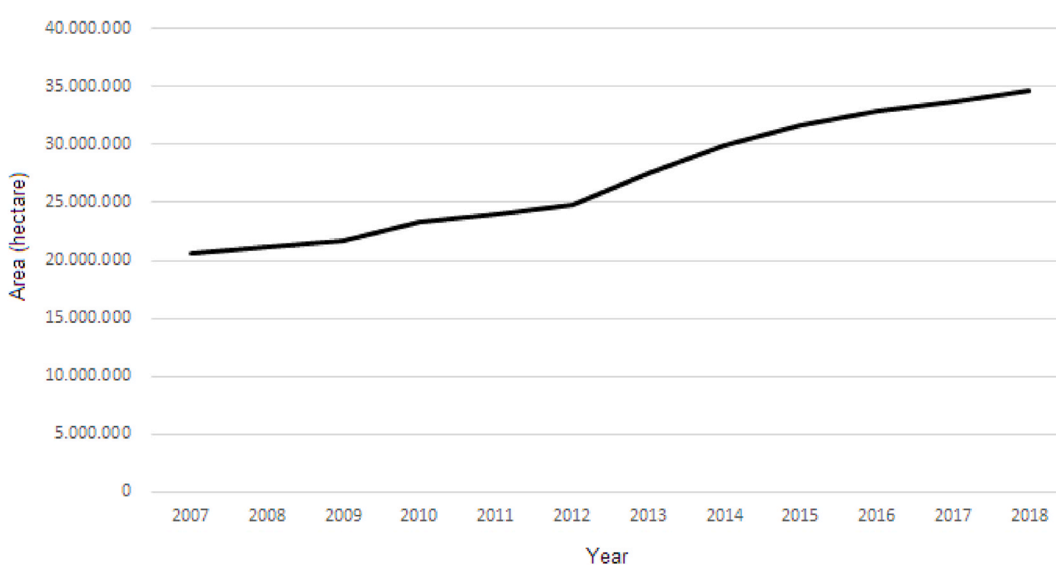

Fig. 8. Soybean cultivated area (in hectares) (IBGE, 2020).

cooperation and coordination of collective actions and no access to official rural credit (due to payment default or inability to meet past debts).

The considerations of Alves et al. (2017) and Conejero et al. (2017) are compatible to the increase in soybean and peanut cultivated areas (Fig. 8), the reduction of castor bean cultivated area and the stable area of sunflower and cotton (Fig. 9). It is worth mentioning that although the peanut cultivated area increased from 2007 to 2018, it is much smaller than the soybean one. It is important to remember that the reduction of castor oil cultivated area, even with a tax reduction seems not to be enough for its use in biodiesel production, mainly due to the advantageous economics in the automobile industry.

Soybean cultivated area increased from 20 million hectares in 2007 to almost 35 million hectares in 2018 (annual increase - 4.8%). Production of the cited beans showed similar behavior to the cultivated areas.

That increase in soybean production put Brazil as one of the most important world players. In 2018, Brazil was responsible for the production of 115 million tons of world production of 362 million tons. That production represents almost 1/3 of world production. From that soybean production of 115 million tons, 83.6 million tons were exported as produced with additional exports of 16.9 million tons of soybean meal and 1.4 million tons of soybean oil. That exports were responsible for an income of more than US$ 40,9 billion (Embrapa, 2020).

Fig. 10 presents the participation of different raw materials in the production of biodiesel, from the beginning. In the early stages of PNPB, data was not trustable, since some of the producers report a mixture of oils. When that difficulty was overpassed, it became clear the predominance of soybean oil in the production of biodiesel. Its participation reached more than 90%. The increase in biodiesel content, biodiesel plant capacity and complexity and the higher prices of soybean oil promoted the increase of the use of animal fats, with participation that stays between 10 and 20%. Although the major fat comes from bovines, fats from chicken and pigs are also used. Cotton oil participation, although important, is less expressive than soybean oil. In the last two years, other materials, such as sunflower oil, peanut oil, castor oil, palm oil (mainly in North and Northeast regions) and used oil (especially in the Southeast region) showed increased participation.

As mentioned by Bergmann et al. (2013), since it is expected that soybean oil remains the most important biodiesel feedstock, in the future, Brazil can improve the economics of soybean agribusiness, increasing the exports of the bran instead of exporting the whole grain.

The increase in the production of other feedstocks with higher oil yield, when compared to soybean, is also expected. Indeed, some researches indicate that feedstock diversification must be studied considering environmental (Rathmann et al., 2012), social (Córdoba et al., 2019), agricultural (Alves et al., 2017) and, economic (Ntaribi and Paul, 2019) concerns since the better results are quite dependent on country or microregion characteristics.

5. Impact on diesel imports

Since we are short on diesel, the use of a mixture of biodiesel on diesel reduced diesel imports by more than 4 million m^3 in 2017 (Fig. 11), which represents almost one-fourth of our diesel imports. From the beginning of the biodiesel program to 2017, it is estimated a reduction of almost 29–30 million m^3. Those values were estimated, considering that biodiesel addition to fossil diesel, at the practiced percents, did not alter significantly its efficiency.

Analyzing Fig. 11, it can be seen that after the world crisis of 2008, diesel imports showed a significant decrease, since diesel demand dropped, and biodiesel use went from 2% to 4% in 2009. In the years 2015 and 2016, diesel imports also showed a reduction, which can be attributed to the shrink of the Brazilian economy, with a reduction of GDP. At that period, biodiesel percent on fossil diesel was fixed at 7%.

It is important to mention that the more important impact on diesel imports is the reduction of the dependency on its import from other countries.

6. After 2018 Scenario

Even with the growth of biodiesel content in diesel, from 2014 to 2018, from 5% to 10%, biodiesel production did not reach the scenario expected by the biodiesel producer market. This disappointment is

Fig. 9. Peanut, Cotton, Sunflower and Castor beans cultivated area (in hectares) (IBGE, 2020).

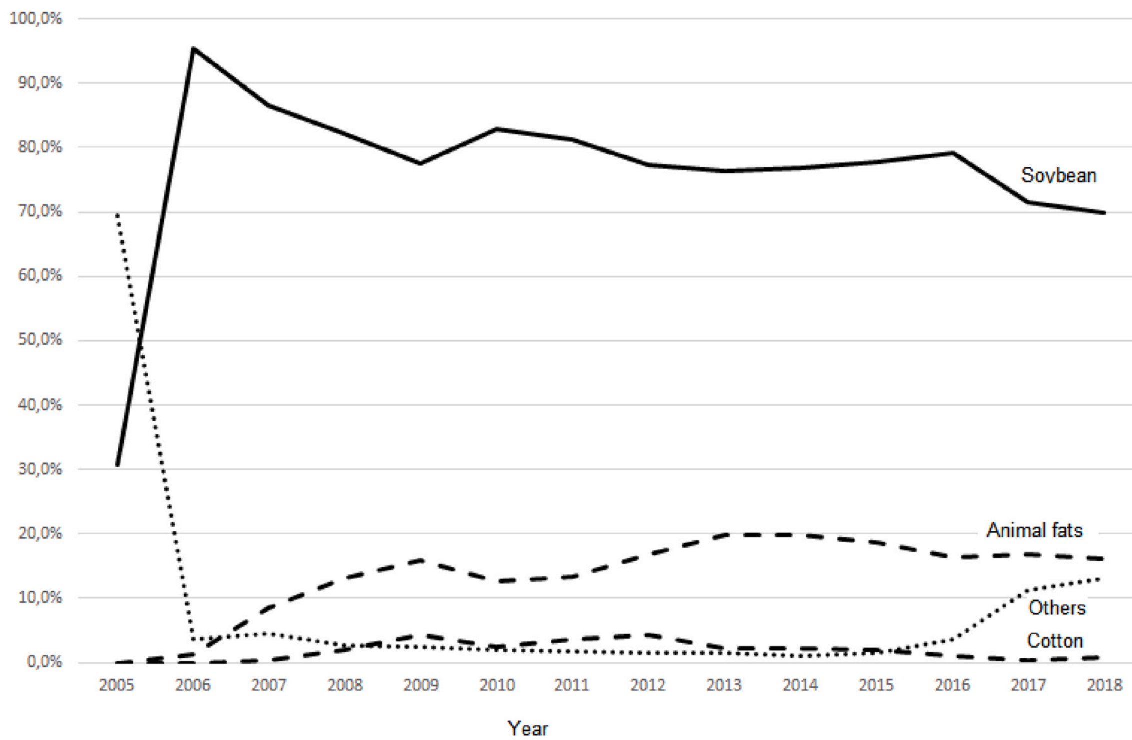

Fig. 10. Participation of soybean, animal fats, cotton and others in biodiesel production (ANP, 2018b).

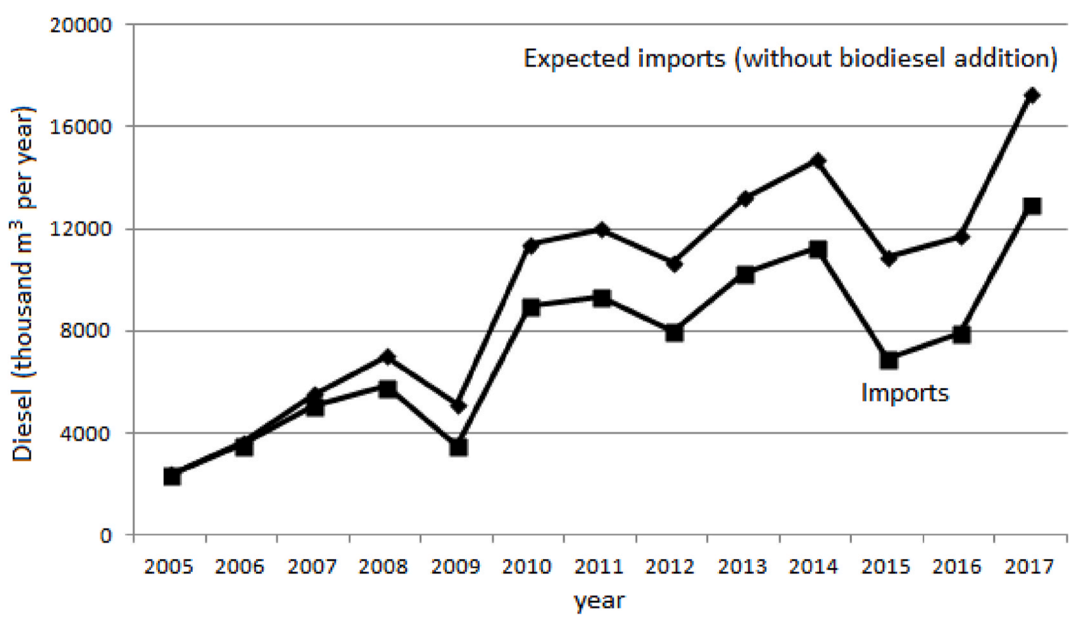

Fig. 11. Diesel imports and estimate of diesel imports without the addition of biodiesel on diesel (ANP, 2018b).

related to the worsening of the Brazilian crisis, with the drop in GDP, leading to a drop in sales of diesel oil (containing biodiesel). Diesel sales in 2014 reached 60.0 billion m^3, with its minimum in 2016, 54.3 billion m^3.

In 2017, biodiesel production was 4.3 million m^3, and it should be remembered that in this year, the percentage of biodiesel increased to 8%. In 2018, with the increase in the biodiesel content to 10%, production increased to 5.4 million m^3 (ANP, 2018b), a value of about 35% higher than the previous year, compatible with the increase in biodiesel content in diesel oil.

Considering the provisions of CNPE Resolution 16 of October/2018, the percentage of biodiesel would rise to 11% on a date subject to previous tests and trials on engines, with a favorable conclusion to an

Table 4
Biodiesel percentual increase.

Biodiesel content (%)	Date
11	September 01, 2019
12	March 01, 2020
13	March 01, 2021
14	March 01, 2022
15	March 01, 2023

increase of up to 15% (fifteen percent). This step was effective on September 01, 2019, as shown in Table 4.

Considering the forecast of a percentage increase in biodiesel and biodiesel production in 2018, we can predict two different scenarios. The first, considering only the projected increase in the content of biodiesel in diesel oil. The second, including the growth of sales of diesel oil by about 2% per annum (equivalent to a GDP growth of 1%) (Table 5).

Considering the forecasts of biodiesel demand, especially for the economic growth scenario (even if moderate), it leads the country to an increase to 9.4 million m^3 of biodiesel in 2024. These volumes will demand that a larger portion of soy be processed in the country, to increase the availability of vegetable oil. Without an increase in soy production, it would be necessary to increase soy processing from 27% in 2018 to around 40–45%.

It is worth mentioning that COVID-19 pandemic impacts are not yet known, but the most expected scenario will be the lower demand scenario, with almost no GDP increase in the period 2020–2024 (strong decrease in 2020 with recovery between 2021 and 2024). As determined by CNPE Resolution 16 of October/2018, on March 1st, 2021, biodiesel content rose from 12% to 13%, but on April 9th, 2021, the CNPE Resolution 4 of April/2021 reduced the biodiesel percent to 10%, for May and June.

Córdoba et al. (2019) studied an important point that can impact future biodiesel production. They studied the palm oil expansions in the Amazon area. The first one took place in the late 1960s and had rotted in an authoritarian and developmental federal government project (Córdoba et al., 2019). The second large oil palm expansion began in the 2000s, boosted by PNPB. It is important that the increase in biodiesel content, together with weak environmental governance that favors agribusiness development, did not take place at the expense of an increase in deforestation and smallholder agriculture.

It is important to mention that Córdoba et al. (2019) used mixed-methods to interpret insights about the impacts of oil palm production. Indeed, both water availability and air and water quality (an increase of agrochemical use) are perceived as the most heavily impacted ecosystem services by the future expansion of this crop. That was potentialized by the perception that extensive changes in land use are expected since the election of 2018 with a permissive vision related to environmental restrictions. This can promote the further expansion of palm oil and other commodities in areas that were previously protected. Indeed, this seems to be confirmed by recent pronouncements of governmental authorities.

That concern is shared by Castanheira et al. (2014) which presented an estimation that the demand for biodiesel until 2020 will need the expansion of oilseed area of more than 6 Mha. That increase in crops related to biofuel production can also be observed by the comparison of the crop's distribution in 1990 and 2010. The increase in soybean and sugarcane contribution was evident.

Castanheira et al. (2014) also pointed their concern on 'Cerrado', that the replacement of small-scale farming and cattle pasture by large-scale soybean monocultures, forcing these small farmers to other regions, such as the Amazon region as well as Rodriguez et al. (2018) studied the impact of soybean production on the same Biome "Cerrado" by analyzing crop water requirements and yield reduction.

A recent article by Andrade et al. (2019) studied some scenarios for future ethanol demand in Brazil, influencing, among others, the land-use competition. To the ethanol industry, the authors concluded that the expansion of sugarcane will lead to minimum deforestation. So, since there are no similar studies on biodiesel, the conclusion reached for ethanol is comforting, but studies must be developed.

7. Conclusion

Biodiesel production rose from almost nothing to more than 4 million cubic meters (at 2017), reaching second place in biodiesel ranking. That was only possible by the enlargement of the biodiesel units, as well as the construction of new ones, bigger than the first ones. In the year 2017, the average capacity is over 10 times the early units in 2005. After a period of stability, the non-competitive biodiesel units went out of the market, with a discrete decrease in the number of operational units.

Table 5
Projection of biodiesel demand.

Year	Scenario 1 (million m^3)	Scenario 2 (million m^3)
2019	5.8	5.9
2020	6.6	6.9
2021	7.2	7.6
2022	7.7	8.4
2023	8.3	9.1
2024	8.4	9.4

During the evolution of the PNPB, the regulation was continuously improved, with the participation of the market and consumers. That made it possible to regulate greater and more complex industrial units (consequently, with higher risks).

Although diversification of biodiesel feedstock was a main objective of PNPB, methanol was the major alcohol used, due to its price, higher conversion and lower recovery costs. When analyzing the other raw material for the production of biodiesel, it is evident that soybean oil and animal fats are responsible for more than 90% of produced biodiesel. The increase in the demand for soybean oil led to an increase in its cultivated area, from 20 million hectares in 2007 to almost 35 million hectares in 2018.

The biodiesel program, besides being an incentive to improve the use of renewable fuels, also improved Brazilian trade balance (imports/exports), by replacing part of diesel that would be necessary for the absence of biodiesel.

The recent events, such as the reduction of oil prices and the pandemic of Covid-19 indicates that the projection made in this article to 2021 will not be reached, since some decrease in biodiesel demand is already seen, but it can be reached in 2024 if the government chose the correct economic tools. Depending on the increase in demand for soybean oil for biodiesel production, it will be necessary the increase in its cultivated area or an increase in the internal processing of soybean, with exports of soybean meal.

Other complementary policies, such as encouraging family farming and the local industry (in the case of Brazil), can be integrated into the biodiesel production policy, with caution and continuous monitoring, so as not to increase the complexity or cost of the program.

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Declarations of interest

None.

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